

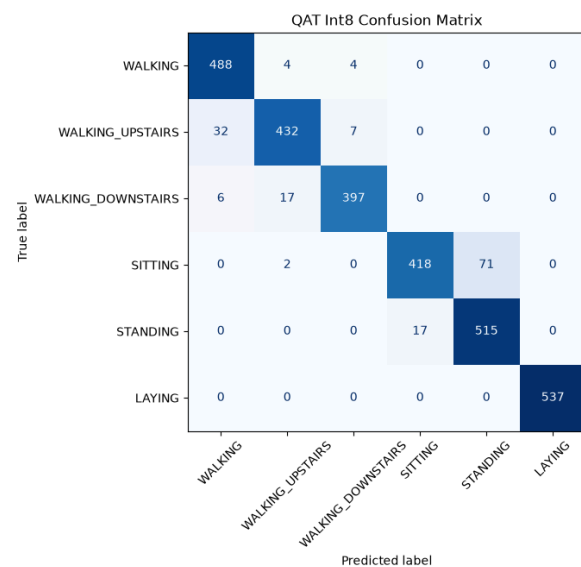
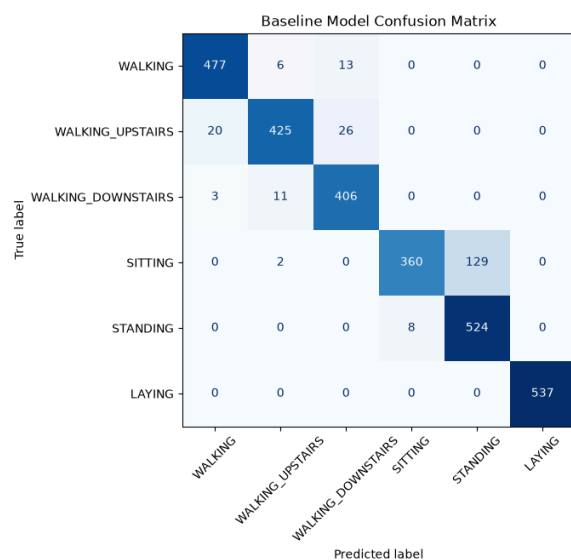
EE 446 Lab 3

Zelamir Reynolds-Henkins

DNN Quantization

PAQ int8 gave the smallest model size but by a very small margin. QAT int8 had the best accuracy of all models, improving upon the accuracy of the PAQ model, and was only slightly larger than the PAQ model. Therefore the QAT model would be the best choice for deployment. The UCI HAR dataset is a good fit for TinyML because it has a small input shape (3 accelerometer readings + 3 gyroscope readings), 6 outputs, and easily quantize-able data.

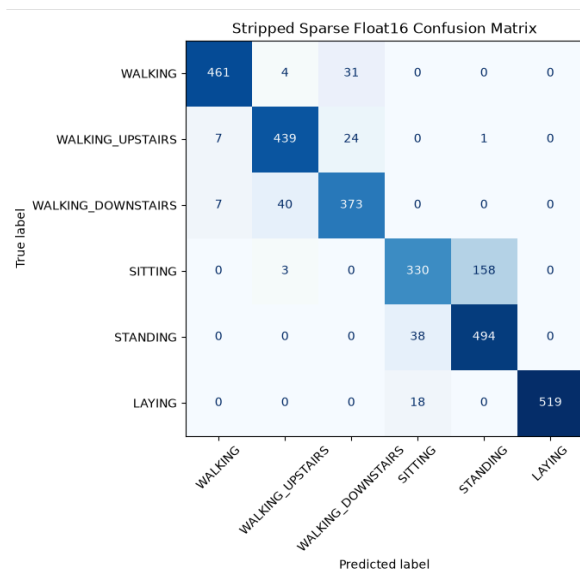
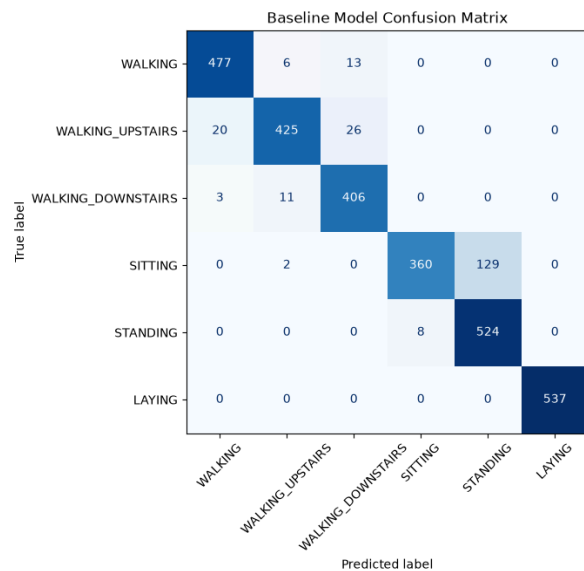
	Model Family	Format	Test Accuracy	Model Size (KB)
0	Baseline	FP32	0.9260	726.75
1	Baseline	Dynamic Range	0.9274	186.14
2	Baseline	Float16	0.9260	365.77
3	Baseline	PAQ INT8	0.9281	185.70
4	Baseline	QAT INT8	0.9457	186.34



DNN Pruning Quantization

Pruning alone actually increased the size of the model because the mask was still integrated. Strip_pruning removed the mask/wrappers which drastically reduced the size of the model. The smallest model was the stripped sparse float 16 model, which remarkably had the same accuracies as the other three pruned models. Because of this, I would choose to deploy this model.

	Model	Format	Test Accuracy	Model Size (KB)
0	Baseline	FP32	0.9260	726.72
1	Pruned	FP32 w/ Mask	0.8877	1454.21
2	Stripped	Sparse FP32	0.8877	227.54
3	Pruned	Float w/ Mask	0.8877	732.62
4	Stripped	Sparse Float16	0.8877	150.94



Knowledge Distillation

This notebook template was a bit disorganized so I struggled to follow some parts. The baseline student and the distilled student have very similar sizes and the distilled student is slightly less accurate. Distillation alone did not help retain performance. After pruning, stripping, and quantization however, it became more accurate and reduced dramatically in size 316kb -> 39kb. The stripped sparse distilled int8 model would make a great candidate for TinyML because it has a very small size and still has ~93% accuracy.

	Model	Test Accuracy	Model Size (KB)
0	Teacher	0.9372	1770.97
1	Student	0.9372	316.85
2	Distilled Student	0.9141	316.89

	Model	Format	Test Accuracy	Model Size (KB)
0	Distilled Student TFLite	FP32	0.914150	316.886719
1	Pruned Distilled TFLite (with mask)	FP32	0.932134	634.687500
2	Stripped Sparse Distilled TFLite	FP32 + Sparse	0.932134	73.699219
3	Stripped Sparse Distilled TFLite	INT8 + Sparse	0.929080	38.875000